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Anticipatory and reflexive neck muscle activities during voluntary rapid jaw opening and passive jaw depression in humans

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SUMMARY **The characteristics of head movement during voluntary rapid jaw opening movement and passive jaw depression were investigated using accelerometers and electromyographs (EMG) on eight healthy examinees. Passive depressions were executed by means of load on the lower jaw, initiated either by examinees themselves or an experimenter. In the depression initiated by examinees, a head-extension movement that preceded the load to the lower jaw and anticipatory activities in the nuchal region of the trapezius muscle were observed. In the depression initiated by the experimenter, the anticipatory activities were not observed. In both of these cases, stretch reflexes were induced in the trapezius muscle. During voluntary rapid jaw opening, a head-extension movement nearly synchronized with the opening movement in the lower jaw acceleration, and dorsal-neck muscle activities accompanying the synchronized movement were observed. The peak timing of these neck-muscle activities preceded the latencies of the stretch–reflex activities observed in the jaw-depressed tasks, but no anticipatory activities were observed in the dorsal-neck muscles. We conclude that neither the anticipatory activities nor the reflex activities observed in the passive depressions have effects on the initial part of the dorsal-neck muscle activities, which are related to the head-extension synchronized with the voluntary lower-jaw opening movement.**

KEYWORDS: **anticipatory postural adjustment, reflex, voluntary movement, head, neck, electromyography**

Introduction

It has been reported that voluntary rhythmical mandibular movements, for example chewing and jaw tapping, are accompanied by synchronized head movements (Arai *et al.*, 1994; Eriksson, Zafar & Nordh, 1998; Yamabe, Yamashita & Fujii, 1999). It has also been reported that cervical muscle activities can be recorded during these head movements (Davies, 1979; Kohno, Yoshida & Kobayashi, 1988; Bérzin, 1995). These reports suggest that there is some relation between the mandibular and the head kinetic systems, but the functional significance of this head movement is not yet fully understood.

Changes in electromyographic (EMG) activity can be recorded in postural muscles of trunk and lower limbs prior to a voluntary focal movement of the upper limbs (Friedli, Hallett & Simon, 1984; Dufossé, Hugon & Massion, 1985; Forget & Lamarre, 1990). These changes have been addressed as anticipatory postural adjustment, and tend to reduce the early perturbations of the body's centre of gravity by shifting the centre in the opposite direction prior to the focal movement (Bouisset & Zattara, 1981). The cervical muscle activities and the head movement during voluntary mandibular movement can be related to this anticipatory postural adjustment. To our knowledge, however, there is only one report concerning the anticipatory postural adjustment of the neck muscles during jaw movement (Nagasuna *et al.*, 1994). Thus, it is not fully understood if the head movement and the neck muscle activities during jaw movement are involved in the anticipatory adjustment. The difference between the neck muscle activities accompanying the voluntary lower jaw

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movement and the sensory induced activities which are induced in the EMGs of dorsal neck muscles by load-stimulation to the stomatognathic system also remains unclear.

We have already reported that head movement during voluntary mandibular movement would not be due to the sensory induced activities of dorsal neck muscles by means of the accelerations of the head and the lower jaw (Torisu *et al.*, 2001).

The aim of the present work is to investigate whether anticipatory postural adjustment and sensory induced activities affect the initial part of the head movement and neck muscle activities accompanying voluntary rapid jaw opening movement.

Materials and methods

Eight healthy examinees (four men, four women; aged 22–30, mean 25·4) volunteered and gave informed consent. The examinees were seated comfortably without a head-rest. Voluntary rapid jaw opening and two kinds of passive jaw depressions in which the loads were initiated in two different ways were compared. The experimental paradigm is shown below.

Voluntary quick open: (Q. open)

Examinees were asked to open the mouth as quickly as possible after an acoustic signal, after holding the intercuspal position while waiting for the signal. A 10-ms square wave amplified by an audio-amp was given as the signal at random time intervals of at least 4 s (Torisu *et al.*, 2001).

Passive jaw depressions

A weight totalling 1·6 kg was used to depress the examinee's lower jaw. This weight was hung at the mandibular premolars with a plastic coated cord 1·5 mm in diameter. Examinees held their mouth slightly open so as not to bite the cord, and were instructed to try not to bite strongly when they were loaded. The load was initiated by one of following two weight-release conditions.

Voluntary weight release (VWR)

Examinees held the weight with their right hands so as not to load on their lower jaws at first. After the acoustic signal, they rapidly released the weight (Fig. 1). The signal condition was the same as for Q. open.

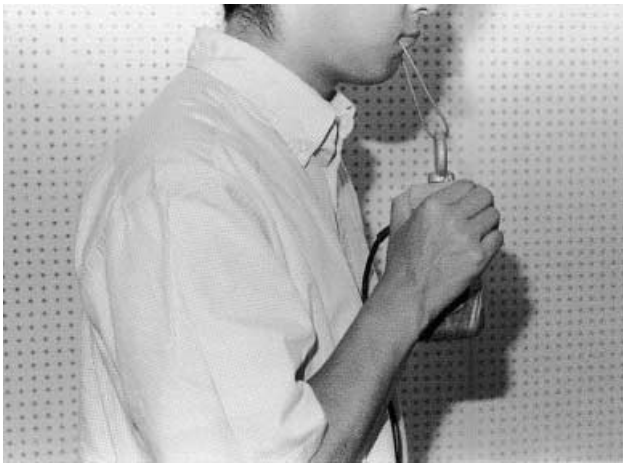

Fig. 1. View of the experimental setup in passive jaw depression initiated by voluntary weight release.

Imposed weight release (IWR)

The experimenter held the weight so as not to load the examinee's jaws at first, and then rapidly released the weight at random time intervals. In this task, the release was started without any prediction to the examinees, and the experimenter took great care so that the examinees could not predict the timing of release using visual and/or auditory sense.

After some practice trials, more than 40 consecutive trials were recorded for each task. These three tasks were performed in random order. The experiments were performed with examinee's eyes open in the two voluntary tasks, and were performed with their eyes closed in the imposed task.

Recording

Vertical accelerations of the head and the lower jaw were recorded by means of piezoelectric accelerometers* (612) and then amplified by charge amplifiers. The accelerometers were fixed rigidly on the labial surfaces of the upper and lower incisors using dental resin, and vertical accelerations in the sagittal plane were picked up to record extension–flexion of the head and open–close of the lower jaw (Fig. 2). Anteroposterior

*TEAC Instruments Co., Kawasaki, Japan.

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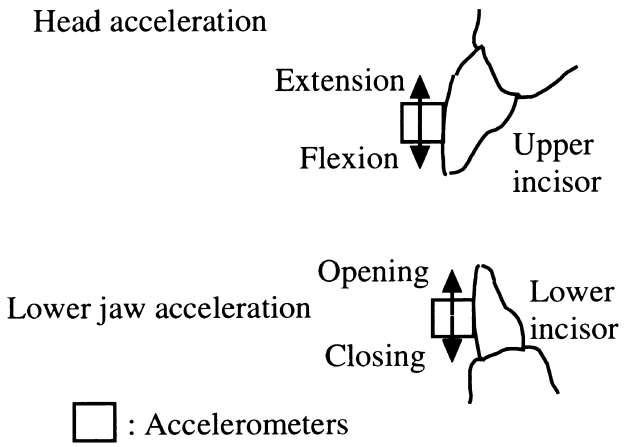

Fig. 2. Location of accelerometers.

trunk-acceleration at the first thoracic vertebra was recorded for six examinees. The depression force of the lower jaw was recorded by a force-transducer attached to the cord hanging the weight.

The surface EMGs were obtained on the left from the masseter muscle (Mm), the anterior digastric (Dig – ant), the insertion of the sternocleidomastoid (SCM), and the nuchal region of the trapezius muscles (Tra-nu), and on the right from the biceps brachii muscle (Bi-bra) using pairs of 8 mm Ag–AgCl disc electrodes 15 mm apart. Fine wire electrodes 80 μm in diameter inserted into the skin 15 mm apart were used for the Tra-nu of five examinees. The EMG-data was recorded simultaneously with the accelerations and the depression force on an FM tape recorder. This data was stored on a personal computer after A–D conversion (MP100†) at a sampling rate of 1500 Hz. Computer software (AcqKnowledge†) was used to the EMGs to accomplish a 30 Hz low cut filter, a 56–64 Hz band-stop filter where necessary, and full-wave rectification.

Analysis

For each examinee and each condition, the EMGs and the accelerations were averaged from 400 ms before to 300 ms after the temporal reference (700 ms total). More than 40 trials were averaged. In Q.open, the onset point of the lower jaw acceleration, which deviated towards the opening direction, was used for the reference to averaging. In the depressed tasks, the onset point of the downward-stimulation estimated in the lower acceleration was used for the reference point. Electromyographs were smoothed for every 10 ms (= 15 points) after averaging. The following variables were measurements on the averaged waves for each examinee and each task.

Acceleration

So as to standardize the measurements, the onset point was defined to be the timing when the amplitude became continuously greater than 120% of the peak-to-peak value of background acceleration. It was checked whether there was a change in the head acceleration that preceded the reference point, and the number of the examinees showing this change was counted. The peak amplitudes and the preceding times from the reference point were also measured with respect to this change. In addition, the number of examinees showing a head acceleration occurring almost simultaneously with the reference point was counted, and the amplitude was measured. The amplitudes of the changes in the head and lower jaw accelerations that occurred immediately after the reference point were also measured.

Electromyography

The anticipatory activities in the jaw muscles have been reported to start about 100 ms before loading (Abbink *et al.*, 1999), and the occurrences of the muscle activities that started about 100 ms before the reference point (Pre100) were therefore checked. The occurrences of the muscle activities that preceded the reference point for about 40 ms (Pre40) were also checked in consideration of the motor time (MT) during voluntary movements (Botwinick & Thompson, 1966). These onset points were defined to be the timing when the amplitude became continuously greater than 120% of the maximum value during the resting period. The maximum amplitudes and the preceding time from the reference point were measured.

The reflex-latency induced by the depressions and the timing of the peak amplitude from the reference point were measured. In VWR, as the muscle activities at the reference point had already increased to greater than that during the resting period, the onset point was defined as the timing when the amplitude became greater than 120% of the minimum value after the reference point.

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Statistics

The paired *t*-test or one-way analysis of variance (ANOVA) was used for testing the significance of the difference between tasks. Differences between individual pairs of groups were compared by means of Fisher's Protected Least Significant Difference (Fisher's PLSD). A significance level of 0·05 was applied in both cases.

Results

Acceleration

The antecedent change in the head acceleration prior to the reference point was observed only in the task at which the load was initiated by examinees themselves (VWR). In this task, this change was observed for all examinees, and the direction was toward head extension. This acceleration change hereafter will be denoted Pre-ext (Fig. 3, Table 1). The preceding time and amplitude were, on average, 90·3 ± 24·8 (mean ± s.d.) ms and 0·12 ± 0·07 *g*, respectively. The Pre-ext was observed in the lower jaw acceleration for all examinees in the VWR, and the preceding time and amplitude were 81·9 ± 21·2 ms and 0·27 ± 0·19 g, respectively. In VWR, the peak amplitude of the anteroposterior trunk-acceleration appeared in the period from 100 ms before, up to the reference point. The value was 0·021 ± 0·031 *g*, and the directions were posterior for five examinees and anterior for one examinee. On the other hand, no Pre-ext was present for all examinees in voluntary quick opening (Q.open, Fig. 3). In IWR, the Pre-ext was observed for only one examinee, and the preceding time was 25 ms (Table 1).

The change towards head extension (Syn-ext) which nearly synchronized with the change in the lower jaw acceleration towards jaw opening was observed for all examinees during head acceleration in Q.open (Table 1). The amplitude was 0·18 ± 0·10 *g*, and was not significantly different from that of Pre-ext in VWR (paired *t*-test, $P = 0{\cdot}199$, Fig. 4). In both jaw-depressed tasks, downward deviations of the head acceleration

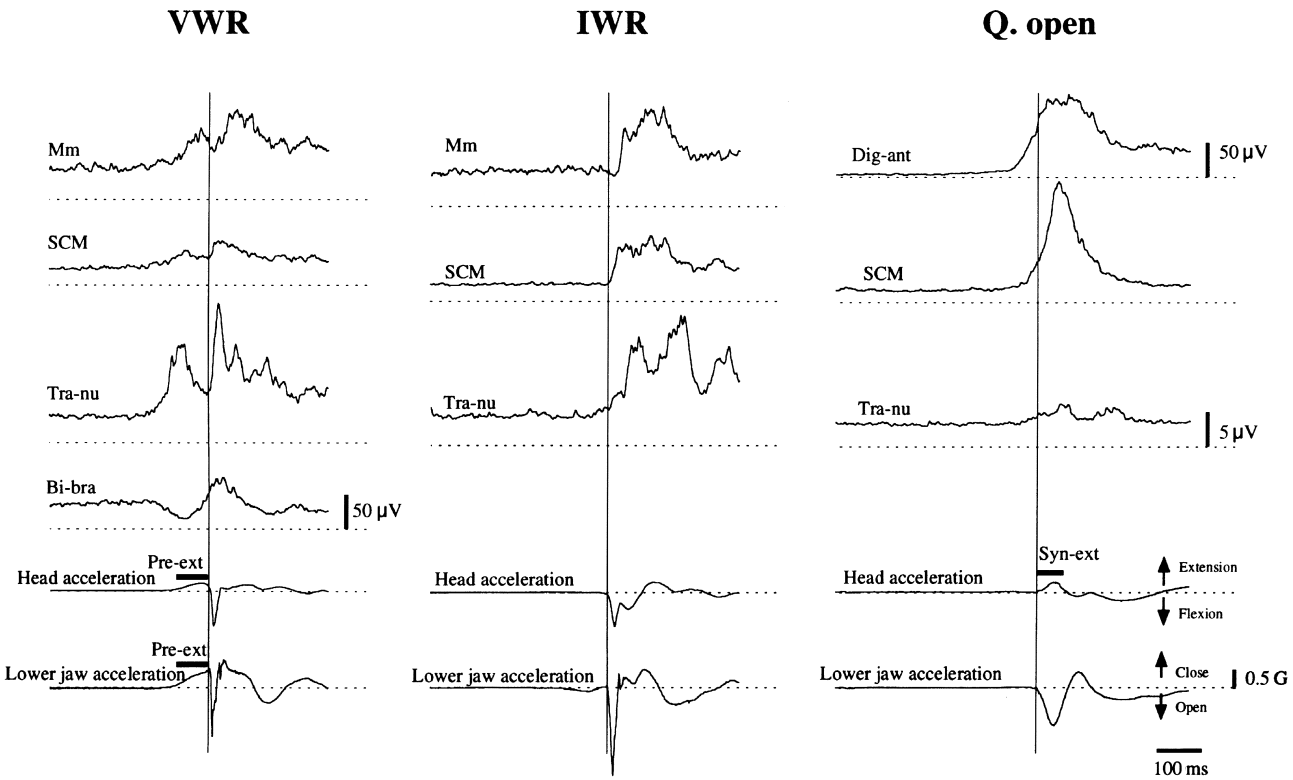

Fig. 3. Full-wave rectified electromyographs (EMGs) and accelerations averaged for at least 40 trials (one examinee). The vertical line presents the onset of downward acceleration at the lower jaw as the reference point. Seven EMG-waves without scale are presented in the same order as in the Tra-nu activity at voluntary quick open. All acceleration-waves are presented in the same order.

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Table 1 Incidences of each change of the head acceleration

Task	Pre-ext	Syn-ext
Q. open	0/8	8/8
VWR	8/8	0/8
IWR	1/8	0/8

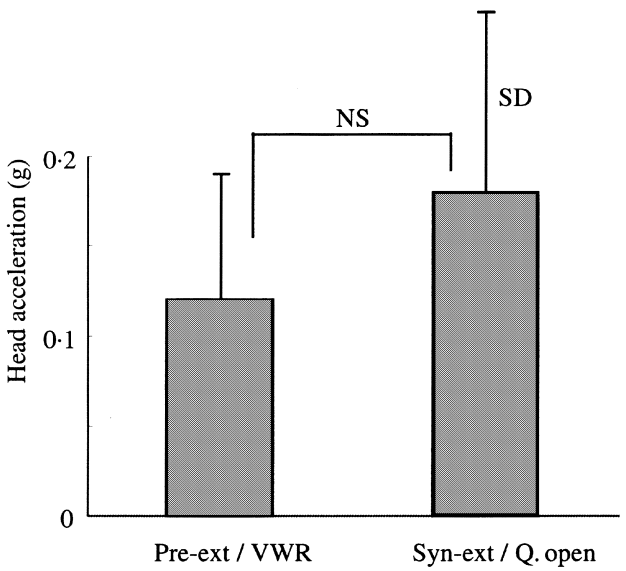

Fig. 4. Mean values of maximum head acceleration. Values are measured on eight averaged wave forms. Directions of both accelerations were towards head–neck extension. $P > 0{\cdot}05$.

were observed after the reference point. The amplitudes were $1{\cdot}08 \pm 0{\cdot}70\ g$ in the VWR and $0{\cdot}85 \pm 0{\cdot}26\ g$ in the IWR, and there was no significant difference between these amplitudes (paired *t*-test, $P = 0{\cdot}463$). The depression forces of the lower jaw were also not significant between the two depressed tasks (VWR: $2{\cdot}76 \pm 1{\cdot}23$ kgw, IWR: $2{\cdot}35 \pm 0{\cdot}69$ kgw).

In the lower jaw acceleration, the changes towards jaw opening caused by the depression were observed after the reference point in both depressed tasks. These amplitudes were $1{\cdot}70 \pm 0{\cdot}72\ g$ in VWR and $2{\cdot}48 \pm 1{\cdot}37\ g$ in IWR. In Q. open, the amplitude of the lower jaw acceleration towards opening was $1{\cdot}03 \pm 0{\cdot}33\ g$. This amplitude was significantly different from that in IWR (Fisher's PLSD, $P < 0{\cdot}01$), but was not different from that in VWR ($P > 0{\cdot}15$) (Fig. 5).

Electromyography

Concerning the neck muscles, Pre100 was observed only in VWR. In Q. open and IWR, Pre100 was not

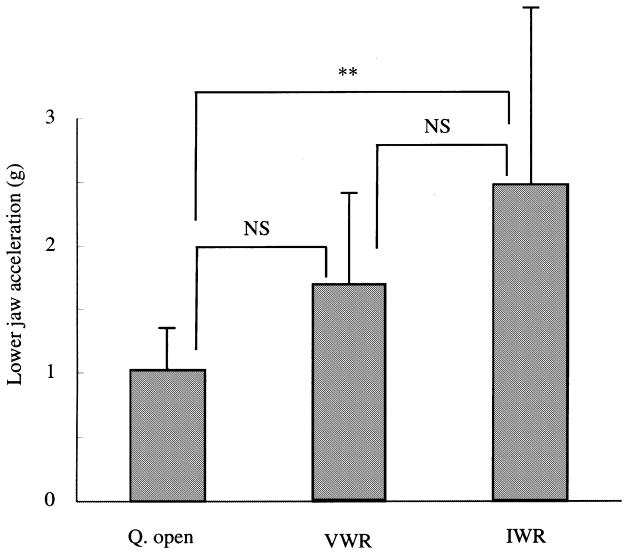

Fig. 5. Mean values of maximum lower jaw acceleration. Values are measured on eight averaged wave forms. Directions of all accelerations were toward mouth opening. $^{**}P < 0{\cdot}01$.

present for any of the examinees (Table 2). For Tra-nu, Pre100 was present for all examinees in VWR; the preceding time was $113{\cdot}8 \pm 33{\cdot}4$ ms, and the maximum amplitude was $15{\cdot}6 \pm 7{\cdot}8$ μV (background activity: $3{\cdot}3 \pm 1{\cdot}0$ μV). For Mm, Pre100 was observed in VWR for five of the eight examinees. The preceding time was $109{\cdot}0 \pm 19{\cdot}2$ ms, and the maximum activity was $48{\cdot}7 \pm 34{\cdot}0$ μV (background activity: $10{\cdot}4 \pm 4{\cdot}3$ μV). Pre100 at Mm was present in Q. open for two examinees, but the maximum activities were low (6·3 μV on average, background activity: 3·8 μV).

Pre40 was observed in Q.open for all examinees at Dig-ant, seven of seven examinees at the insertion of SCM, and six of eight examinees at Tra-nu (Table 2). The preceding times were $49{\cdot}3 \pm 14{\cdot}3$, $38{\cdot}4 \pm 15{\cdot}8$ and $43{\cdot}2 \pm 15{\cdot}7$ ms, the maximum activities were $140{\cdot}7 \pm 38{\cdot}2$ (background activity: $3{\cdot}7 \pm 1{\cdot}5$ μV), $15{\cdot}5 \pm 6{\cdot}8$ (background activity: $2{\cdot}0 \pm 0{\cdot}7$ μV), and $17{\cdot}3 \pm 5{\cdot}3$ μV (background activity: $2{\cdot}5 \pm 1{\cdot}0$ μV),

Table 2. Incidences of each change in muscle activities

	Pre100			Pre40		
Task	Mm	SCM	Tra-nu	Dig-ant	SCM	Tra-nu
Q.open	2/8	0/7	0/8	8/8	7/7	6/8
VWR	5/8	2/7	8/8	–	0/7	0/8
IWR	0/8	0/7	0/8	–	1/7	0/8

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Table 3. Latencies and timings of peak amplitudes for induced activities at Tra-nu. (ms)

Task	Latency	Peak timing
Q.open	–	5·8 ± 10·6 (Pre40)
VWR	9·5 ± 5·1	25·8 ± 12·6
IWR	30·6 ± 13·4	56·3 ± 7·0

For only Q.open, the timing of the peak amplitude was indicated at about Pre40. The latencies were compared by means of the paried *t*-test, and the peak timings were compared by means of ANOVA. $^{**}P < 0{\cdot}01$.

respectively. No Pre40 was present in the jaw-depressed tasks except for one examinee at SCM in IWR.

In both of the jaw-depressed tasks, reflex-activities with latencies of 9·5 and 30·6 ms were observed at Tra-nu for all examinees. The latency was significantly shorter in VWR than in IWR (paired *t*-test, $P < 0{\cdot}01$). The peak timing of Pre40 in Q.open was significantly shorter than those of the reflex-activities in both jaw-depressed tasks (Fisher's PLSD, $P < 0{\cdot}01$) (Table 3).

Discussion

It has been reported that anticipatory postural activities occur in the distal-to-proximal order in arm movements (Friedli, Hallett & Simon, 1984). Accordingly, trunk movements might affect our results. However, the value of the trunk acceleration in VWR before the reference point was less than about one-sixth of those at the head and the lower jaw, and the directions in anteroposterior were not constant. Nevertheless the directions of the head acceleration in this period were toward head extension for all examinees. Thus, trunk movement is not thought to substantially affect the head and lower jaw acceleration in this study.

The head extension (Pre-ext) and neck muscle activities (Pre100) before the reference point observed in VWR were not found in IWR. These findings indicate that these preceding changes were not evoked by the other mechanical stimuli accompanying loading, for example a stimulus to the skin around the lips and/or a change in the atmosphere surrounding the examinees, but by the examinees' anticipation of the actual loading to the lower jaw.

Anticipatory postural adjustment appears to minimize the postural perturbation caused by voluntary movement and loading (Hugon, Massion & Wiesendanger, 1982). In the voluntary quick opening task of this study, no anticipatory head movement was observed. In this task, the change of the head acceleration occurred almost simultaneously and it was in the opposite direction from that of the lower jaw. Moreover, the onset of neck muscle activities occurred after the onset of Dig-ant, acting as the prime mover for opening movement (Pre40). On the other hand, there was no difference between VWR and the voluntary opening in the amplitudes of the lower jaw accelerations toward opening. Nevertheless, in VWR, the anticipatory head movement was prior to the depression of the lower jaw. The preceding times in the muscle activities of Mm and Tra-nu in VWR were about 110 ms, and these values were in accord with those obtained by Abbink *et al.* (1999). These results suggest that the anticipatory postural adjustment is not related to the head movement which is observed during the voluntary lower jaw movement in the context of the initial period of this movement.

In this study, the anticipatory activities of Mm in VWR were not observed in some examinees. It has been reported that jaw closing muscles tend to respond to loads mainly through reflex loops rather than anticipatory response during rhythmic jaw movements, even when loading of the jaw is anticipated (Abbink *et al.*, 1999). They have also indicated that this is different from the upper arm muscles, which tend to resist the loading chiefly by means of a large anticipatory response. Thus, our result in Mm is consistent with their findings. On the other hand, in the dorsal neck muscle, anticipatory activities were observed for all examinees. This suggests that the dorsal neck muscle has similar characteristics to those of the upper arms, as opposed to those of the jaw closing muscles. Nevertheless, in the voluntary quick opening, anticipatory activities of the dorsal neck muscle were not observed. These results may also indicate there is little possibility that the head movement observed during the voluntary quick opening is related to the anticipatory activities.

In reaction-time tasks, anticipatory activity has been reported to occur later in time by observation of upper arm movements, typically simultaneously with the initiation of the focal movement as compared with that in the self-paced movement (Wolf, Slijper & Latash, 1998; Nougier, Teasdale, Bard & Fleury, 1999). These researches suggest that coordination of posture and focal movement may be triggered by a simultaneous and unique central command. In the work reported here, the voluntary quick opening and the head

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movement accompanying the lower jaw movement may be triggered by the same central command; the anticipatory activity may not be obvious prior to the focal movement, and it may be merely hard to distinguish. It may have occurred nonetheless, however, because the voluntary quick opening in this study was a reaction time task.

Motor time (Botwinick & Thompson, 1966) was shown to be 30–50 ms for the jaw muscles and the neck muscles in our previous paper (Torisu *et al.*, 2001). The Pre40 observed in this study preceded the onset of the lower jaw movement for about 40 ms. Furthermore, the Pre40 was observed only in the voluntary opening movement except for one examinee in IWR. Accordingly, the Pre40 is interpreted as motor time to jaw-opening.

Trigemino-cervical reflex and/or the stretch reflex of the dorsal neck muscles may be related to the induced muscle activities observed after the stimulation, because the stimulation in this study was given to the area innervated by the trigeminal nerve and was a stimulation to stretch the dorsal neck region.

It has reported that stimulation of the area supplied by the trigeminal nerve elicits a short-latency reflex in dorsal neck muscles (Manni *et al.*, 1975; Sumino, Nozaki & Katoh, 1981). The reflex has been known as trigemino-cervical reflex, and the latency has been reported to be 13–35 ms for SCM and trapezius in humans (Nakashima *et al.*, 1989; Lazzaro *et al.*, 1995). These latencies coincide with that of the induced muscle activities in this study. However, Lazzaro *et al.*, (1995) reported that the stimulation to the facial skin supplied by the infraorbital nerve was the most effective, while stimuli to the mental nerve did not produce clear effects. The stimulation in this study was given to the mandible, so our stimulation might be ineffective in terms of inducing the trigemino-cervical reflex. The induced activities observed after the stimulation at the depressed tasks would thus be mainly because of a different reflex from the trigemino-cervical reflex, that is the stretch reflex, although we cannot negate the involvement of the trigemino-cervical reflex.

It has been demonstrated that the latency of the jaw jerk reflex becomes shorter during voluntary contraction (Goldberg, 1972). Also in our study, the latency of the induced muscle activities at Tra-nu was shorter in VWR than in IWR. But in both of the two tasks, the latencies were longer than previous research on the latency of the H-reflex and the stretch reflex (about 7 ms) for Mm and the temporal muscle in humans (Fujii & Mitani, 1973; Fujii, 1977). In the depressed tasks of this study, the stretch reflex of the dorsal neck muscles was evoked not by direct stimulation to the neck muscles but through the stimulation to the lower jaw. Therefore, these longer latencies may be related to insufficient extension of dorsal neck-muscle spindles (Abbink *et al.*, 1999), or to insufficient voluntary contraction of the dorsal neck muscles before loading.

As mentioned, in the voluntary quick opening, dorsal neck muscle activity that began about 40 ms before the jaw opening was observed. The timing when this activity reached a peak was earlier than the peak timings and the onsets of the stretch–reflex activities in the jaw depressed tasks. The peak timing in the voluntary opening was also prior to the latency of the trigemino-cervical reflex. From these findings, the reflex activities would not affect the initial stage of the head movement observed during the voluntary jaw opening movement.

We conclude that the head movement and the lower jaw movements would be controlled by tight-knit central commands or the same command (Nougier *et al.*, 1999).

Acknowledgments

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